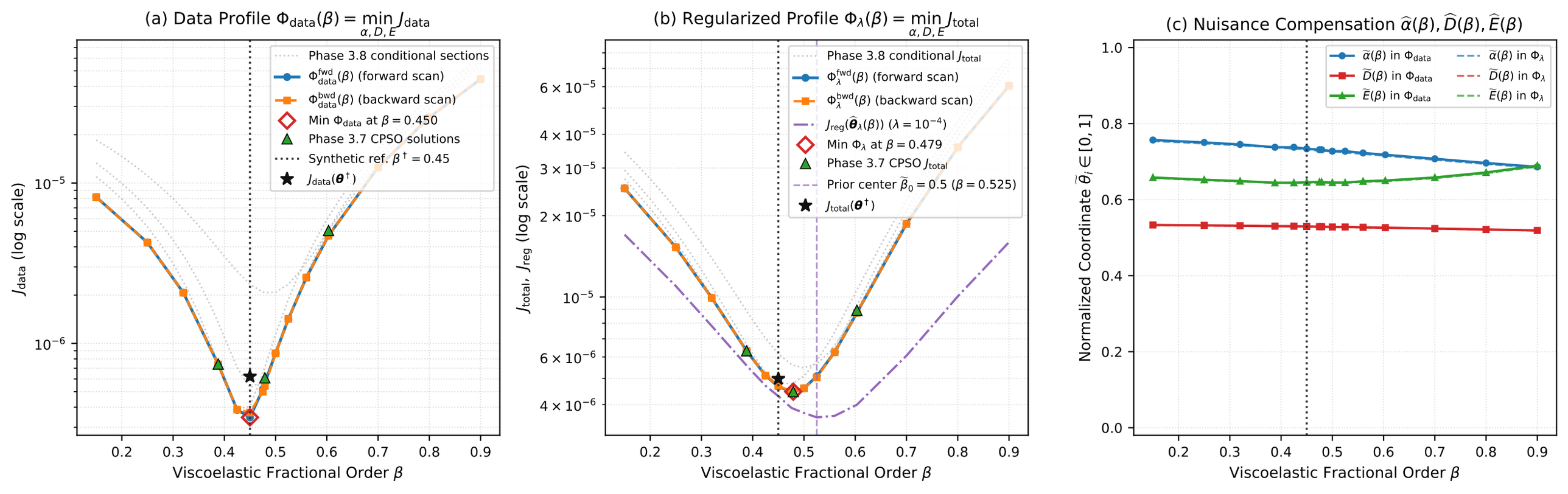
(a) Data Profile $\Phi_{\text{data}}(\beta) = \min_{\alpha, D, E} J_{\text{data}}$
Phase 3.8 conditional sections
$\Phi^{\text{fwd}}_{\text{data}}(\beta)$ (forward scan)
$\Phi^{\text{bwd}}_{\text{data}}(\beta)$ (backward scan)
Min Φ_{data} at $\beta = 0.450$
Phase 3.7 CPSO solutions
Synthetic ref. $\beta^\dagger = 0.45$
$J_{\text{data}}(\boldsymbol{\theta}^\dagger)$
J_{data} (log scale)
Viscoelastic Fractional Order β
(b) Regularized Profile $\Phi_\lambda(\beta) = \min_{\alpha, D, E} J_{\text{total}}$
Phase 3.8 conditional J_{total}
$\Phi^{\text{fwd}}_\lambda(\beta)$ (forward scan)
$\Phi^{\text{bwd}}_\lambda(\beta)$ (backward scan)
$J_{\text{reg}}(\widehat{\boldsymbol{\theta}}_\lambda(\beta))$ ($\lambda = 10^{-4}$)
Min Φ_λ at $\beta = 0.479$
Phase 3.7 CPSO J_{total}
Prior center $\widetilde{\beta}_0 = 0.5$ ($\beta = 0.525$)
$J_{\text{total}}(\boldsymbol{\theta}^\dagger)$
J_{total}, J_{reg} (log scale)
Viscoelastic Fractional Order β
(c) Nuisance Compensation $\widehat{\alpha}(\beta), \widehat{D}(\beta), \widehat{E}(\beta)$
$\widetilde{\alpha}(\beta)$ in Φ_{data}
$\widetilde{D}(\beta)$ in Φ_{data}
$\widetilde{E}(\beta)$ in Φ_{data}
$\widetilde{\alpha}(\beta)$ in Φ_λ
$\widetilde{D}(\beta)$ in Φ_λ
$\widetilde{E}(\beta)$ in Φ_λ
Normalized Coordinate $\widetilde{\theta}_i \in [0, 1]$
Viscoelastic Fractional Order β